

Online Appendix

Price Regulation in Credit Markets: A Trade-off between Consumer Protection and Credit Access

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A Robustness Exercises and Additional Results

The empirical strategy developed in Section 3.2.2 exploits policy variation across time and loan-size brackets to measure the effect of interest rate caps on credit market outcomes. The main concern with this strategy is the potential for equilibrium effects on loans larger than \$8,000, which are not directly treated by the policy change, but might be affected through some form of substitution along that margin. In this section, we implement different robustness exercises that provide support to our empirical strategy. Moreover, we include some additional results.

A.1 Substitution across Loan Size Brackets

Given regulation becomes stronger for loans of \$0-\$8,000, consumers could substitute towards that group, which would imply our quantity effects are attenuated. However, differences across interest rate caps limit such incentives, as the cap for loans over \$8,000 is lower than that for loans of \$0-\$8,000 throughout, as shown in Figure 1. On the other hand, banks may offer multiple loans below \$8,000 rather than a single loan larger than \$8,000 to charge higher interest rates. However, we find no evidence of such behavior in the data, as more than 98% of borrowers take only one loan in months in which they borrow, and that share remains unchanged throughout, as displayed by Figure A.10.

A.2 Effects of Placebo Policies

Our approach in Section 3.2.2 relies on a comparison across treated and untreated loan-size brackets. One should not expect to find the same estimated effects across different comparison groups within untreated loan size brackets. We study whether that is the case, by estimating equation (4) for placebo policies. Concretely, we use the same definition for the policy and the same policy intensity variables as in Section 3.2.2 to estimate effects of interest rate regulation on different parts of the loan size distribution. In practice, we proceed by replacing the dependent variable y_{krt} to $y_{k+\Delta,rt}$, where Δ defines the placebo policy. We start by policy size brackets defined as being \$8,000 higher than actual ones, and then sequentially increase them by \$2,000 to generate a range of placebo policies.

Figure A.11 displays the results from this exercise for price and quantity outcomes. Each figure displays our main estimates from Table 3 and Table 4, along with estimates for a range of placebo

policies. Figures A.11-a and A.11-b display results for maximum and average interest rates, and the results are stark: estimates from placebo policies are remarkably different from our estimates and close to zero. Figures A.11-c and A.11-d display results for quantity outcomes, for which placebo estimates are noisier but offer a similar pattern: most of the point estimates are close to zero and not statistically different from zero. These results provide evidence against particular patterns of substitution from untreated loan size brackets to treated loan size brackets.

A.3 Effects on Distribution of Application Loan Size and Term

An additional robustness exercise we implement is to study the evolution of the distribution of application amount and term. If there substitution across policy size brackets, that should be reflected in a change in the distribution of application loan amount. Figure A.12-a shows the evolution of relevant statistics of the distribution of application loan amount separately for loans smaller than \$8,000 and loans larger than \$8,000. We find no evidence of changes in this distribution around the date of the policy change we study. Moreover, the same is true for application loan term, as displayed by Figure A.12-b.

A.4 Effects around Policy Thresholds for Loan Size

The approach proposed in Section 3.2.2 exploits loans larger than \$8,000 as a control group for evaluating the effect of the policy. One concern regarding that is that, in response to the change in relative interest rate regulation between loans under and above that threshold, there could be equilibrium effects affecting loans larger than \$8,000 despite them not being directly treated by the reform we study. For example, one could argue that, before the reform, borrowers could have sought to get loans right above the \$8,000 threshold to benefit from lower interest rate caps on loans in that loan-size bracket than on loans of amounts marginally below such threshold. That incentive would be reduced by the policy, given interest rate caps for both groups were brought closer by it. In such case, we should observe bunching at that threshold from above before the policy, and a decrease in such behavior after it. That in turn would imply that our results in Section 3.2.2 would underestimate the effects of the reform. On the other hand, banks may have opposite incentives to induce borrowers to take marginally smaller loans on the left side of the policy threshold or to take multiple small loan instead of a large one, for which we already provided evidence against in Figure A.10. This incentive would also decrease under stronger regulation.

In order to address this concern about substitution around the threshold, we study the distribution of the number of loans around relevant policy thresholds. Figure A.13 displays the number of loan originations at loan sizes around relevant policy thresholds, before and after the policy. As displayed by Figure A.13-a, The relationship between the density of loan size below the \$2,000 threshold is remarkably noisy—this pattern is driven by mass points in the loan size distribution that are observed at certain round number for loan size—, which makes it difficult to conclude

anything. However, above the \$2,000 threshold, there is no noticeable change in such density before and after the policy. A similar comparison is displayed by Figure A.13-b for the \$8,000 policy threshold. While the distribution of the number of loans shows more mass around the policy threshold after the policy, that behavior is similar on both sides of threshold.¹

In a more systematic attempt to address this concern, we repeat the analysis in Section 3.2.2 dropping loans around the \$8,000 policy threshold. Table A.3 displays results from estimating equation (4) excluding loans of size between \$6,000 and \$10,000 from the sample. Estimates are quantitatively similar to those obtained with the full sample in Section 3.2.2, which is reassuring in terms of our empirical strategy. Finally, Figure A.14 repeats this exercise for price and quantity outcomes for a variety of comparison groups which differ in their lower bound, and provides evidence in the same direction: estimates for the effects of the policy do not change substantially when excluding loans close to the policy threshold from the comparison group.

A.5 Alternative Comparison Groups

Our analysis in Section 3.2.2 exploits loans between \$8,000 and \$20,000 as a comparison group for those directly affected by the policy change. In this subsection, we assess how our estimates would change under alternative definitions of compares groups. In particular, we estimate the same specification as in (4) but for variety of comparison groups, starting with loans between \$8,000 and \$10,000, and then increasing sequential by \$2,000 until a group covering loans between \$8,000 and \$30,000.

Results from this exercise are displayed in Figure A.15, and show results for price and quantity outcomes. Each figure displays our main estimates from Table 3 and Table 4, along with estimates for a range of alternative comparison groups. Overall, the main conclusions of our main analysis are unaffected, as point estimates do not change substantially across comparison groups. On the other hand, there are some efficiency gains from using larger comparison groups, which reflects in tighter standard errors.

A.6 Heterogeneity across Banks

We have focused so far on the effects of interest rate regulation at the market level. In this subsection, we provide results for heterogeneity in effects across banks. Figure A.16 provides results for marginal effects of interest rate regulation on both number of lines and prices for each of the 8 largest banks in the market. While there is heterogeneity in magnitudes, our estimates suggest that the stronger interest rate regulation affects most banks in the same direction, by inducing them to sign less loan contracts and to do so at lower interest rates. This is consistent

¹Figures A.13-c and A.13-d complement this analysis by showing that average interest rates shift downwards after the policy, but that there is no discontinuity in average interest rates around policy thresholds. We should mention that when looking at high enough percentiles in the distribution of loan interest, discontinuities at the policy thresholds become evident, which is consistent with bunching at the interest rate cap displayed in Figure 2.

with our estimates for average effects and with the interpretation we give to them.

Interestingly, there is 1 bank that displays a different behavior, by reducing average interest but simultaneously increasing credit volume as a result of stronger interest rate regulation. Those estimates suggest that either borrowers substituted towards that bank which perhaps had a more lenient screening process or that the bank changed its screening process as a result of the policy change.

B Estimation Details

B.1 Preliminary Steps in Estimation of Application Equation

We discuss joint estimation of the application and repayment equations in (10) and (11) in Section 5.3. Estimation proceeds in three steps, of which the first two are related to estimation of components of the application equation in equation (10) that are not observed for every consumer in the sample, and that we then use as inputs in estimation of the key parameters in that equation by maximum likelihood. We provide further detail about those steps in this section.

In the first step, we estimate the conditional distribution of loan amount and term (L_i, T_i) using data from applicants and then draw from that distribution for non-applicants. In the first stage, we estimate a probit model for applications on a rich vector of borrower covariates x_i that includes the level and change of consumer and mortgage debt and default, income, consumer and mortgage debt to income ratio, age and gender; and application shifters z_i that include the total number of branches across banks in the consumer local market and the lagged number of related banks:

$$P(a_i = 1) = \Phi(x_i' \beta_a + z_i' \gamma_a)$$

which we estimate this model on data for the period before the policy change in December 2013.

In the second step, we estimate a regression of loan amount on the same vector of borrower covariates, and include fitted propensity scores $\hat{P}(x_i, z_i)$ as a control function to account for selection into application. This procedure is based on Das *et al.* (2003) and also used by Attanasio *et al.* (2008) for studying loan demand. In particular, we estimate:

$$\ln(L_i) = x_i' \beta_L + \lambda(\hat{P}(x_i, z_i)) + \epsilon_i$$

where λ is the control function.

In the third step, we estimate an ordered logit model for bins of months of loan term on borrower covariates and loan size:

$$\begin{aligned} P(T_i = j) &= P(\alpha_{j-1} < x_i' \beta_T + \delta_T L_i \leq \alpha_j) \\ &= \Lambda(\alpha_j - (x_i' \beta_T + \delta_T L_i)) - \Lambda(\alpha_j - (x_i' \beta_T + \delta_T L_i)) \end{aligned}$$

where α_{j-1} and α_j are the cutoffs that define loan term monthly bin j . The advantage of using an ordered logit model in this step is that it accommodates the fact that the empirical distribution of loan term features noticeable spikes at multiples of semesters. Using estimates from these regressions, we first take draws of loan amount for consumers that did not apply for loans $\tilde{L}_i$, conditional on x_i ; and then take draws of loan term for consumers that did not apply for loans $\tilde{T}_i$, conditional on x_i and L_i .

In the second step, we estimate the approval probability P_C and the density of loan prices conditional on approval $f_{p|C}$. We estimate P_C using a probit model for an indicator of application approval on a vector of borrower covariates, previous relationship variables, as well as application amount and term from the first step and month fixed and market effects:

$$P(C_i = 1) = \Phi(x_i' \eta + \zeta_t + \tau_m)$$

which we estimate separately for low- and high-risk borrower to allow for flexibility. We include time and market fixed effects in order to accommodate the possibility that approval probabilities change over time and across markets due to the policy change. We compute expected approval probabilities $\hat{P}_{Ci}$ as the fitted values from this equation for each consumer in the sample. Then, we estimate the density of loan monthly payments conditional on approval $f_{p|C}$, using a kernel density estimator after conditioning on the same vector of variables. We then take draws from this estimated conditional density for estimation of borrower preferences by maximum likelihood below.

B.2 Likelihood Function for Application and Repayment

The parameters of interest in the application equation are those in v_{Ci} , δ_p and κ_i , whereas the parameters of interest in the repayment equation are α_S . Moreover, we are interested in the parameters in the joint distribution of application and repayment shocks, namely ρ and σ_S . Recall that σ_A is normalized to 1.

We start by the likelihoods of application choices. Given the normality assumption we impose on ε_{Ai} , the probabilities that a potential borrower chooses to apply and not to apply are:

$$\begin{aligned} P_{a_i=1} &= \Pr \left(P_{Ci} \int (x_i' \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp - z_i' \kappa + \varepsilon_{Ai} > 0 \right) \\ &= \Phi \left(P_{Ci} \int (x_i' \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp - z_i' \kappa \right) \\ P_{a_i=0} &= \Pr \left(P_{Ci} \int (x_i' \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp - z_i' \kappa + \varepsilon_{Ai} < 0 \right) \\ &= \Phi \left(-P_{Ci} \int (x_i' \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp + z_i' \kappa \right) \end{aligned}$$

where Φ is the standard normal cdf. We approximate the integral through simulation by taking $N_A^S = 100$ Halton draws from the estimated conditional density of loan monthly payments conditional on approval, which is:

$$\int (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp \approx \frac{\sum_{s=1}^{N_A^S} (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p_i^{(s)}) 1\{L_i^{(s)} = 1\}}{\sum_{s=1}^{N_A^S} 1\{L_i^{(s)} = 1\}}$$

where $L_i^{(s)}$ is an indicator for whether a loan application by borrower i in simulation draw s was approved.

We now derive the likelihoods of repayment outcomes. The likelihood of observing a given repayment behavior can be written in terms of the distribution of ε_{Si} conditional on application, for which we exploit the properties of conditional normal distributions. There are three cases of interest, one in which borrower i fully repays, one in which borrower i partially repays, and one in which borrower i does not repay at all. The probabilities for these three cases are:

$$\begin{aligned} P_{S=0|a_i=1} &= \Pr(\exp(x'_i \alpha_S + \varepsilon_{Si}) \leq \chi | a_i = 1) \\ &= \int_{-P_{Ci}}^{\infty} \int (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp + z'_i \kappa \Phi\left(\frac{\ln(\chi) - x'_i \alpha_S - \mu_{S|Ai}}{\sigma_{S|A}}\right) f_{\varepsilon_A}(\varepsilon_A) d\varepsilon_A \\ P_{S=s_i|a_i=1} &= \Pr(\exp(x'_i \alpha_S + \varepsilon_{Si}) = s_i | a_i = 1) \\ &= \int_{-P_{Ci}}^{\infty} \int (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp + z'_i \kappa \frac{1}{\sigma_{S|A}} \phi\left(\frac{\ln(s_i) - x'_i \alpha_S - \mu_{S|Ai}}{\sigma_{S|A}}\right) f_{\varepsilon_A}(\varepsilon_A) d\varepsilon_A \\ P_{S=1|a_i=1} &= \Pr(\exp(x'_i \alpha_S + \varepsilon_{Si}) \geq 1 | a_i = 1) \\ &= \int_{-P_{Ci}}^{\infty} \int (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp + z'_i \kappa \Phi\left(\frac{x'_i \alpha_S + \mu_{S|Ai}}{\sigma_{S|A}}\right) f_{\varepsilon_A}(\varepsilon_A) d\varepsilon_A \end{aligned}$$

where $\chi = \frac{1}{T_i} \approx 0$ is the repayment share achieved after the first payment on the contract, and ϕ is the standard normal pdf. Given the joint normality assumption of $(\varepsilon_{Ai}, \varepsilon_{Si})$, the conditional distribution of ε_S is given by:

$$\varepsilon_S | \varepsilon_A \sim N\left(\frac{\rho \sigma_S}{\sigma_A} \varepsilon_A, \sigma_S^2 (1 - \rho^2)\right)$$

which we use for computing the integral in each likelihood by simulation. In particular, we take 100 Halton draws $\varepsilon_{Ai}^{(s)}$ from its truncated marginal distribution, and then compute the conditional mean of ε_{Si} for each draw, $\mu_{S|Ai}^{(s)} = \frac{\rho \sigma_S}{\sigma_A} \varepsilon_{Ai}^{(s)}$.

The likelihood of an observation varies across five observed combinations of application and repayment events, which we indicate using I_i^1 through I_i^5 . The log likelihood of the data is:

$$\begin{aligned} \log \mathcal{L}^D &= \frac{1}{N} \sum_i I_{1i} \log P_{a_i=0} + I_{2i} \log P_{a_i=1} + I_{3i} [\log P_{a_i=1} + \log P_{S=0|a_i=1}] \\ &\quad + I_{4i} [\log P_{a_i=1} + \log P_{S=s_i|a_i=1}] + I_{5i} [\log P_{a_i=1} + \log P_{S=1|a_i=1}] \end{aligned}$$

where I_{1i} indicates non-applicates, I_{2i} indicates rejected applicants, I_{3i} indicates applicants that do not repay at all, I_{4i} indicates applicants that only partially repay, and finally I_{5i} indicates applicants that fully repay. This log likelihood can be written as:

$$\begin{aligned} \log \mathcal{L}^D = & \frac{1}{N} \sum_i I_{1i} \log P_{a_i=0} + (I_{2i} + I_{3i} + I_{4i} + I_{5i}) \log P_{a_i=1} \\ & + I_{3i} \left[\log \int_{-P_{Ci}}^{\infty} \int_{\int (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp + z'_i \kappa}^{\infty} F_{\varepsilon_D|\varepsilon_A}(\ln(\chi) - x'_i \alpha_S | \varepsilon_A) f_{\varepsilon_A}(\varepsilon_A) d\varepsilon_A \right] \\ & + I_{4i} \left[\log \int_{-P_{Ci}}^{\infty} \int_{\int (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp + z'_i \kappa}^{\infty} f_{\varepsilon_D|\varepsilon_A}(\ln(s_i) - x'_i \alpha_S | \varepsilon_A) f_{\varepsilon_A}(\varepsilon_A) d\varepsilon_A \right] \\ & + I_{5i} \left[\log \int_{-P_{Ci}}^{\infty} \int_{\int (x'_i \delta_X + \delta_L L_i + \delta_T T_i - \delta_p p) f_{p|C}(p) dp + z'_i \kappa}^{\infty} F_{\varepsilon_D|\varepsilon_A}(x'_i \alpha_S | \varepsilon_A) f_{\varepsilon_A}(\varepsilon_A) d\varepsilon_A \right] \end{aligned}$$

We maximize this log-likelihood to estimate the parameters of interest in the application and repayment equations.

B.3 Likelihood Function for Bank Costs

The parameters of interest on the supply side are $\{\tau, \gamma, \sigma_\omega\}$. The likelihood function for prices and loan application outcomes can be obtained using results for the distributions of order statistics of the T1EV distribution assumed for ω_{ij} , as detailed in Appendix B.4. We work separately on the corresponding likelihood for each of the three potential outcomes generated by the model and stated in equation (8). In each case, we derive the likelihood conditional on a consideration set $\mathcal{J}_i$, and then integrate across consideration sets using their distribution.

Likelihood for Unconstrained Loans. When regulation is not binding, loan price is the optimal unconstrained price for the lowest cost bank. The likelihood of a contract in such situation, signed at price $p_i \leq \bar{p}_i$ with bank b_i , is:

$$\begin{aligned} P(P_i = p_i, P_i < \bar{p}, B_i = b_i | \mathcal{J}_i) &= P(P_i^u = p_i, P_i^u < \bar{p}_i, B_i = b_i | \mathcal{J}_i) \\ &= \left(g_{i(1)} \left(\frac{f_i - \frac{E_\varepsilon[\varphi(S_i)]}{\varphi(T_i)} p_i}{l_i} | \mathcal{J}_{i \setminus j} \right) \right. \\ &\quad \left. + (\rho_{ib_i}(\mathcal{J}_i) - 1) g_{i(1)} \left(\frac{f_i - \frac{E_\varepsilon[\varphi(S_i)]}{\varphi(T_i)} p_i}{l_i} | \mathcal{J}_i \right) \right) 1(P_i < \bar{p}) \end{aligned}$$

where $g_{i(1)}(\omega)$ is the density of the first order statistic of match-value and $\rho_{ib_i}(\mathcal{J}_i)$ is the choice probability of the bank chosen by borrower i .

Likelihood for Constrained Loans. When regulation is binding and a loan is approved, loan price equals the price cap and the lowest cost bank makes profits from the loan. In such a scenario,

loan price is less than the optimal unconstrained price but higher than the cost of the lowest cost bank. The likelihood of a contract in such situation, signed at price $p_i = \bar{p}_i$ with bank b_i is:

$$\begin{aligned}
P(P_i = \bar{p}_i, B_i = b_i | \mathcal{J}_i) &= P\left(P_i^u > \bar{p}, \frac{\varphi(T_i)}{E_\varepsilon[\varphi(S_i)]}(f_i - l_i \omega_{i(1)}) \leq \bar{p}_i, B_i = b_i | \mathcal{J}_i\right) \\
&= \left(G_{i(1)}\left(\frac{f_i - \frac{E_\varepsilon[\varphi(S_i)]}{\varphi(T_i)} \bar{p}_i}{l_i} | \mathcal{J}_i \setminus j\right) + (\rho_{ib_i}(\mathcal{J}_i) - 1) G_{i(1)}\left(\frac{f_i - \frac{E_\varepsilon[\varphi(S_i)]}{\varphi(T_i)} \bar{p}_i}{l_i} | \mathcal{J}_i\right) \right) \\
&\quad \times \left(1 - G_{i(1)}\left(\frac{f_i - \frac{E_\varepsilon[\varphi(S_i)]}{\varphi(T_i)} \bar{p}_i}{l_i} | B_i = b_i, \mathcal{J}_i\right) \right)
\end{aligned}$$

where $G_{i(1)}(\omega)$ is the distribution of the first order statistic of match-value.

Likelihood for Rejected Loans. When a loan is rejected, the lowest cost bank does not make any profit out of it. Therefore, the cost of such bank is higher than the price cap on the loan. The likelihood of a contract in such situation is:

$$\begin{aligned}
P(P_i = \cdot, B_i = \cdot | \mathcal{J}_i, r_i) &= P\left(\frac{\varphi(T_i)}{E_\varepsilon[\varphi(S_i)]}(f_i - l_i \omega_{i(1)}) > \bar{p}_i | \mathcal{J}_i\right) \\
&= G_{i(1)}\left(\frac{f_i - \frac{E_\varepsilon[\varphi(S_i)]}{\varphi(T_i)} \bar{p}_i}{l_i} | \mathcal{J}_i\right)
\end{aligned}$$

Likelihood Function. The likelihood of the data combines the individual likelihoods for these three cases. Let I_i^u , I_i^c and I_i^r indicate that the outcome for application by i was an unconstrained approved loan, a constrained approved loan or a rejected application respectively. The full log-likelihood function for the data is:

$$\begin{aligned}
\log \mathcal{L}^S &= \sum_{i \in \mathcal{A}} I_i^u \log \left[\sum_{\mathcal{J}_i \in \mathcal{C}_i} P(P_i = p_i, P_i < \bar{p}, B_i = b_i | \mathcal{J}_i, x_i) P(\mathcal{J}_i) \right] \\
&\quad + I_i^c \log \left[\sum_{\mathcal{J}_i \in \mathcal{C}_i} P(P_i = \bar{p}, B_i = b_i | \mathcal{J}_i, x_i) P(\mathcal{J}_i) \right] + I_i^r \log \left[\sum_{\mathcal{J}_i \in \mathcal{C}_i} P(P_i = \cdot, B_i = \cdot | \mathcal{J}_i, x_i) P(\mathcal{J}_i) \right]
\end{aligned}$$

where $\mathcal{C}_i$ is the set of potential consideration sets, and $P(\mathcal{J}_i)$ is the probability if each consideration set. These probabilities are known at this point, given the consideration set model is estimated before estimating bank costs. The number of potential consideration sets in $\mathcal{C}_i$ is $2^{12} = 4,096$. To reduce the dimension of the integration step in the likelihood function above, we instead maximize a simulated log likelihood function by taking N_C draws of consideration sets $\tilde{\mathcal{J}}_i$ from

the distribution of consideration sets, which we collect in set $\tilde{\mathcal{C}}_i$:

$$\begin{aligned} \log \mathcal{L}^S = & \sum_{i \in \mathcal{A}} I_i^u \log \left[\frac{1}{N_C} \sum_{\tilde{\mathcal{J}}_i \in \tilde{\mathcal{C}}_i} P(P_i = p_i, P_i < \bar{p}, B_i = b_i | \tilde{\mathcal{J}}_i, x_i) \right] \\ & + I_i^c \log \left[\frac{1}{N_C} \sum_{\tilde{\mathcal{J}}_i \in \tilde{\mathcal{C}}_i} P(P_i = \bar{p}, B_i = b_i | \tilde{\mathcal{J}}_i, x_i) \right] + I_i^r \log \left[\frac{1}{N_C} \sum_{\tilde{\mathcal{J}}_i \in \tilde{\mathcal{C}}_i} P(P_i = \cdot, B_i = \cdot | \tilde{\mathcal{J}}_i, x_i) \right] \end{aligned}$$

B.4 Useful Properties of T1EV Distribution

In this Appendix, we summarize useful properties of the T1EV distribution, which we use in the derivation of the likelihood function. Proofs for these results are available in [Froeb *et al.* \(1998\)](#). First, it can be shown using the properties of extreme value distributions, that the cdf of the highest utility across banks for a borrower is given by:

$$G_{(1)}(\omega; \mathcal{J}_i) = G(\omega; (\omega_{i,max}, \sigma_\omega), \mathcal{J}_i)$$

where:

$$\omega_{i,max} = \sigma_\omega \log \sum_{j \in \mathcal{J}} \exp \left(\frac{\delta_{ij}}{\sigma_\omega} \right)$$

is the location parameter in the distribution.

Moreover, the probability that j is the bank with the lowest cost for i among those in $\mathcal{J}_i$ is given by the usual logit formula:

$$\rho_{ij} \equiv P(u_{ij} = \max_{k \in \mathcal{J}_i} u_{ik}; \mathcal{J}_i) = \frac{\exp \left(\frac{\delta_{ij}}{\sigma_\omega} \right)}{\sum_{k \in \mathcal{J}} \exp \left(\frac{\delta_{ik}}{\sigma_\omega} \right)}$$

Finally, we can also derive an analytical expression for the distribution of the second order statistic of ω_{ij} in terms of that of the first order statistic. Conditional on j being the choice of borrower i , such distribution would be:

$$G_{(2)}(\omega | u_{ij} = \max_{k \in \mathcal{J}_i} u_{ik}; \mathcal{J}_i) = \frac{1}{\rho_{ij}} G_{(1)}(\omega; \mathcal{J}_{i \setminus j}) + \frac{\rho_{ij} - 1}{\rho_{ij}} G_{(1)}(\omega; \mathcal{J}_i)$$

which by integrating over j to recover the unconditional distribution of the second order statistic yields:

$$\begin{aligned} G_{(2)}(\omega; \mathcal{J}_i) &= \sum_{j \in \mathcal{J}_i} \rho_{ij} G_{(2)}(\omega | u_{ij} = \max_{k \in \mathcal{J}} u_{ik}; \mathcal{J}_i) \\ &= \sum_{j \in \mathcal{J}_i} G_{(1)}(\omega; \mathcal{J}_{i \setminus j}) + G_{(1)}(\omega; \mathcal{J}_i)(1 - |\mathcal{J}_i|) \end{aligned}$$

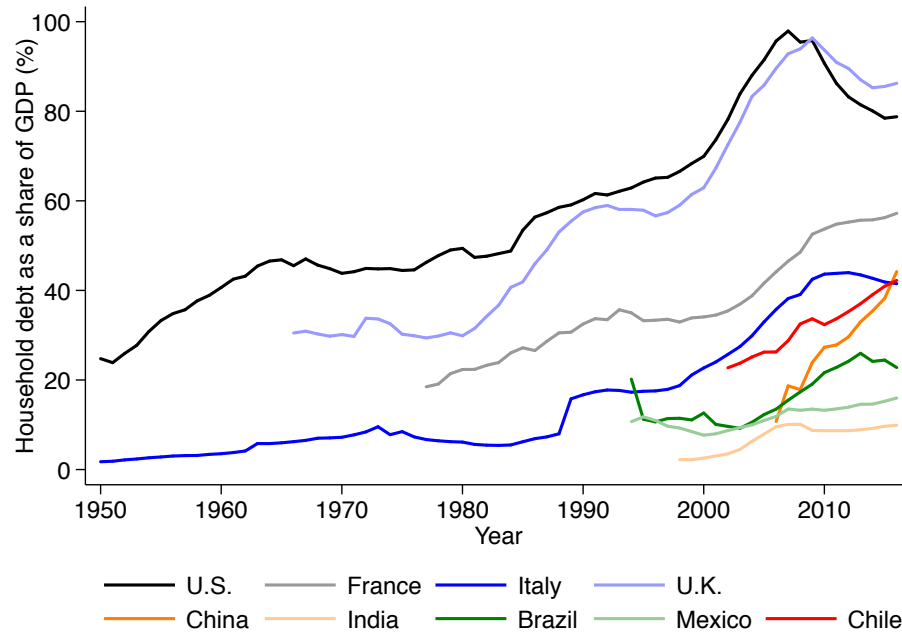
B.5 Robustness Checks of Application Equation Estimates

As discussed in Section 5.2, a concern for our strategy is the potential for unobservables that drive both application decisions by borrowers and pricing decisions by banks, which would influence our estimates of price sensitivity, δ_p . We evaluate this concern by adopting a control function approach to provide evidence for the robustness of $\hat{\delta}_p$. We follow Petrin and Train (2010) and implement a two-step procedure. In the first step, we regress monthly payments on covariates in x_i and a cost shifter of prices. We use funding cost as a shifter, where the funding rate provides variation across time, and heterogeneity in loan amount and term across consumers provides individual level variation. In the second step, we include the residuals from the first stage as an additional covariate in x_i , with the objective of controlling for unobservable drivers of prices. Our estimate for the coefficient on the control function is 4.207 with standard error 1.282. This suggests that including it indeed controls for such potential unobservables. However, the fact that estimates $\hat{\delta}_p$ remain similar to those without the control function suggests that the relative importance of those unobservables relative to observables in (x_i, L_i, T_i) is minor. In particular, estimates in this case $\hat{\delta}_p$ using this approach are 0.713 and 0.488 for low- and high-risk borrowers, whereas estimates from our baseline specification are 0.719 and 0.465, respectively. While not conclusive, these results suggest that the set of covariates in x_i included in estimation might be able to deal with the concern about unobservable drivers of applications and pricing.

References

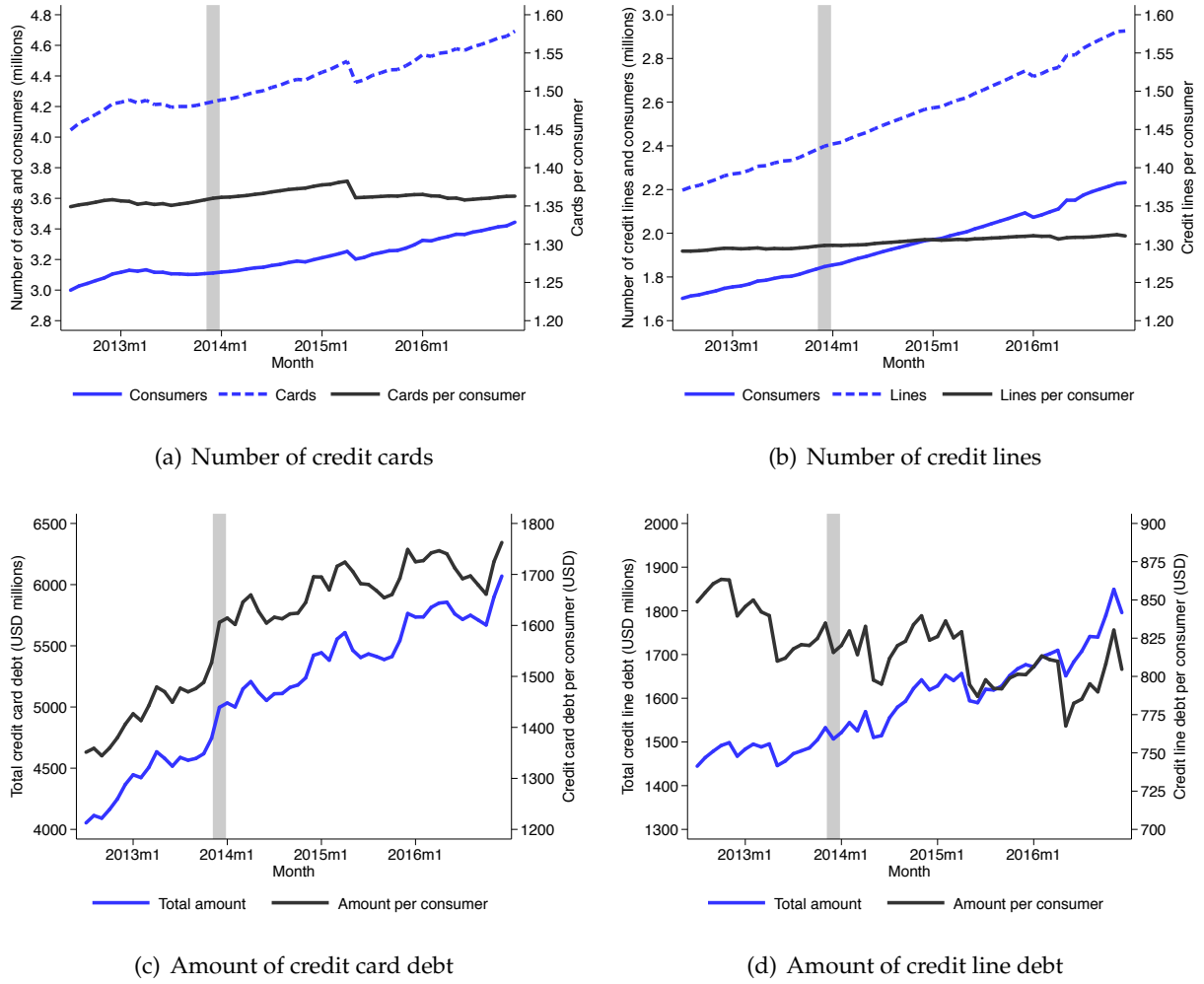
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Figure A.1: Evolution of household debt as a share of GDP across time and countries



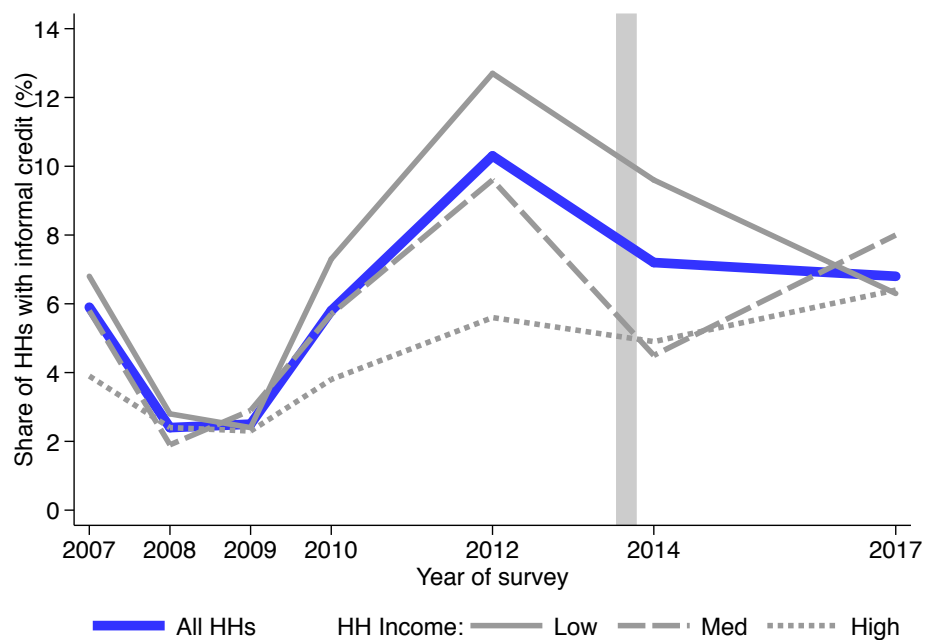
Notes: This figure displays the evolution of household debt as a share of GDP for a sample of countries. Authors' calculation based on data from the Global Debt Database by the International Monetary Fund ([Mbaye et al., 2018](#)). The length of each series is determined by the availability of data for each country. There is substantial cross sectional variation. For 2016, household debt as a share of GDP ranges from 0.6% for Afghanistan to 126.3% for Switzerland, with an average of 39.7%. Calculations in Section 1 are based on a sample of 82 developed and developing countries with available data for 2016.

Figure A.2: Evolution of credit card and credit line debt



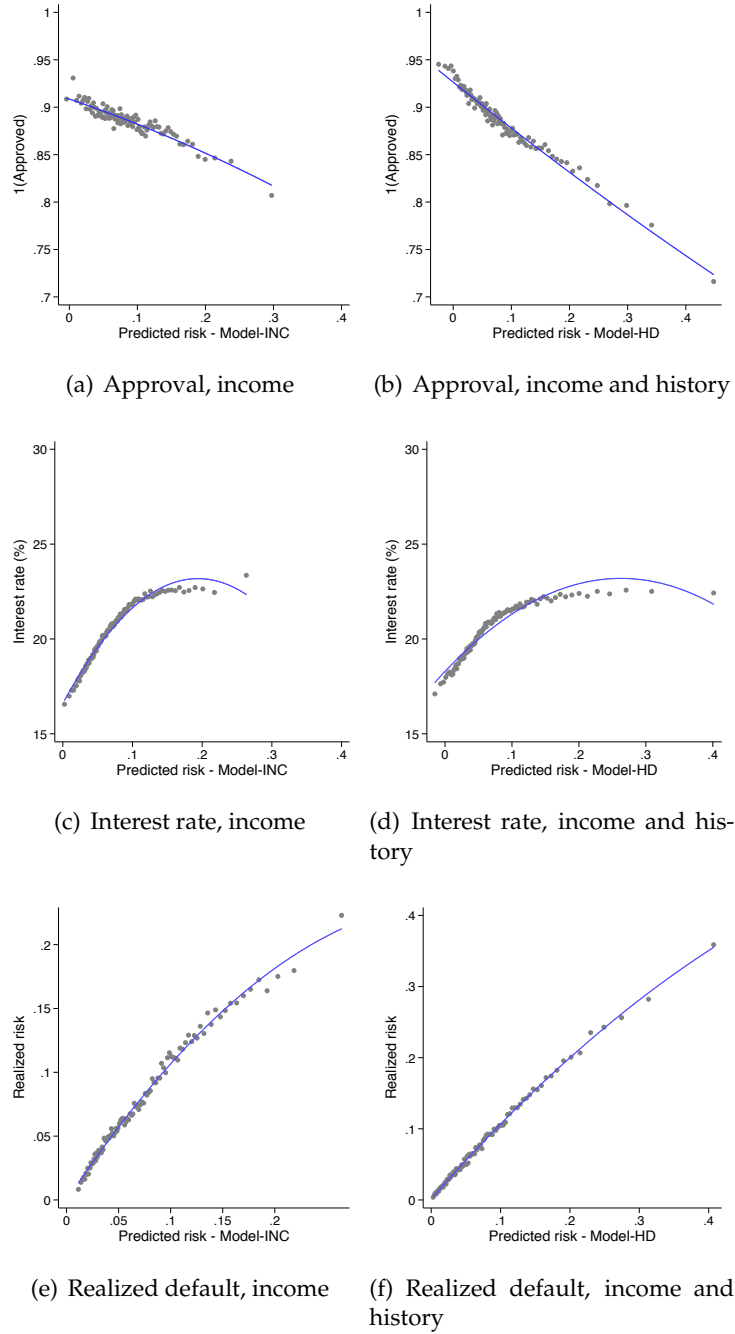
Notes: This figure displays the evolution of the the markets for credit cards and credit lines in the Chilean market using administrative data from CMF. Panels (a) and (b) display the number of consumers and cards/lines (blue) as well as the number of cards/lines per consumer in the market (black). Panels (c) and (d) display the total amount of credit card/line debt in the market (blue) and per consumer in the market (black). The vertical gray line indicates the policy change.

Figure A.3: Evolution of informal debt penetration



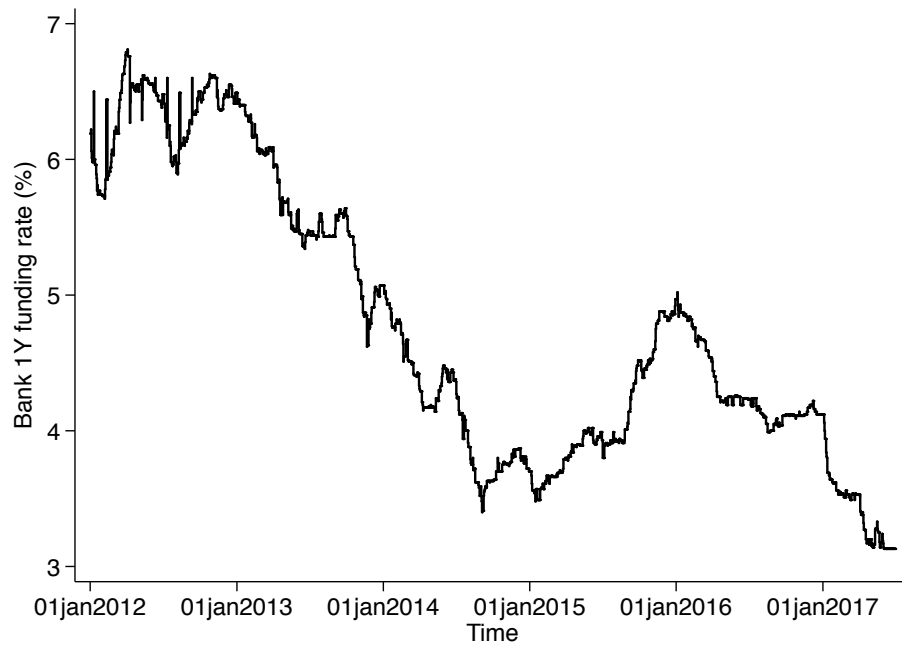
Notes: This figure displays the evolution of the share of households in the Chilean market that hold some kind of informal debt, as measured by the nationally representative Survey of Consumer Finance (EFH, 2018). This statistic covers several sources of informal credit, including family and friends, informal lenders, pawn shops, among others. Data is only available for selected years displayed in the x-axis, which are the ones in which the survey has been implemented. The blue line measures the overall share, whereas gray lines measure shares by terciles of household income. The vertical gray line indicates the policy change.

Figure A.4: Predicted and realized default



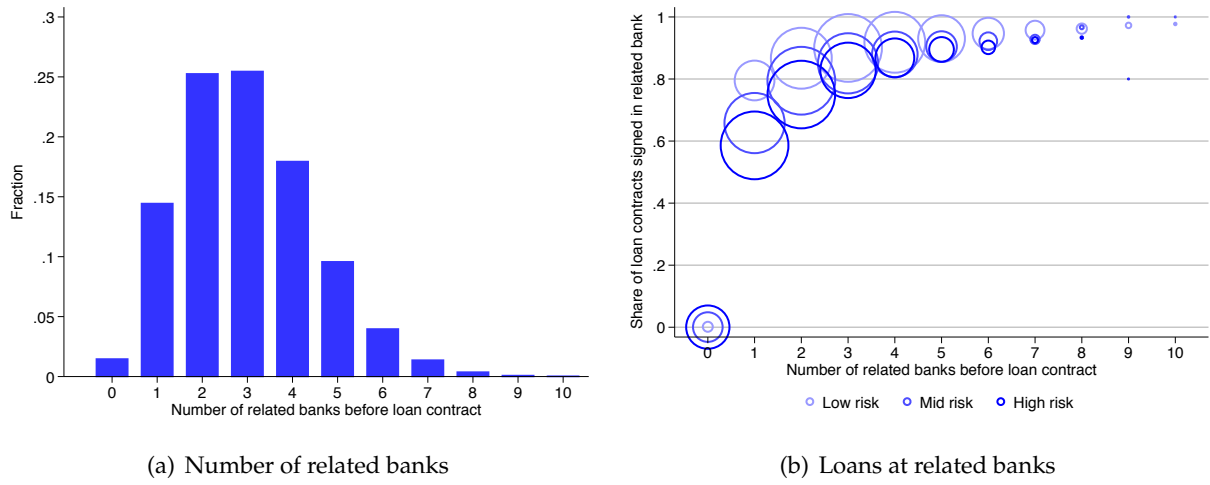
Notes: This figure displays binned scatterplot of predicted loan default probability as constructed using the model described in Section 2.2 and realized outcomes. The left column displays results using borrower income and loan to income ratios as main predictors of default, while the right columns adds a long vector of credit history covariates. Panels (a) and (b) display the relationship between predicted default and loan application approval; Panels (c) and (d) display the relationship between predicted default and interest rate; while Panels (e) and (f) display the relationship between predicted default and realized default. Each dot measures average realized default for loans in each of 100 quantiles of predicted default. The blue line is a quadratic fit of the relationship between both variables.

Figure A.5: Evolution of the funding cost of banks



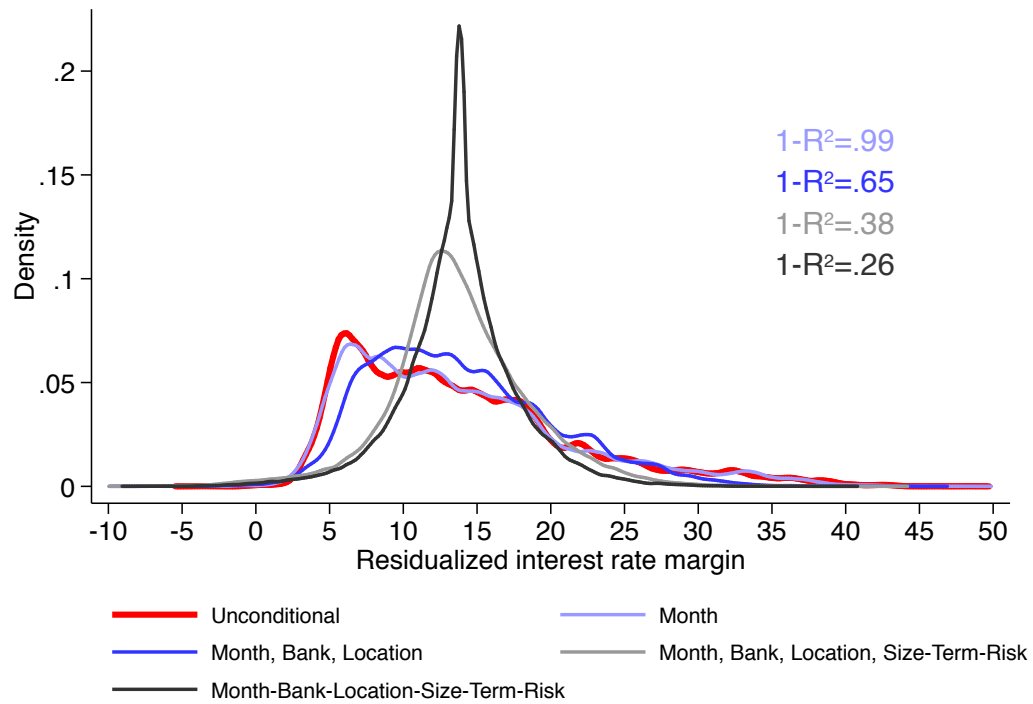
Notes: This figure displays the evolution of the funding cost of banks. This funding cost is calculated as a weighted average of banks deposit rates.

Figure A.6: Relationship between borrowers and banks



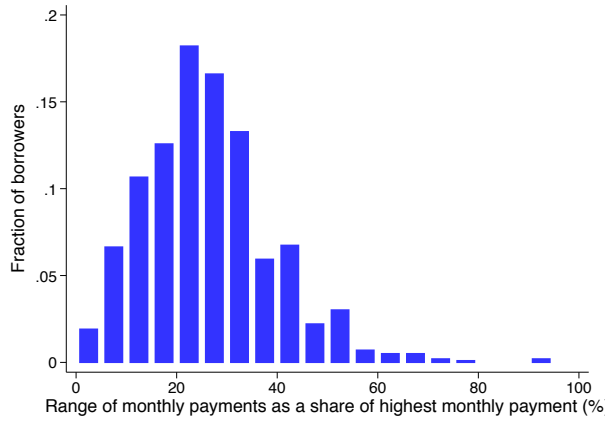
Notes: Panel (a) describes the share of previously related banks for each borrower in the dataset. Panel (b) describes the share of loan contracts signed with a previously related bank for each tercile of borrower risk and each number of previously related banks.

Figure A.7: Price dispersion in consumer loan contracts

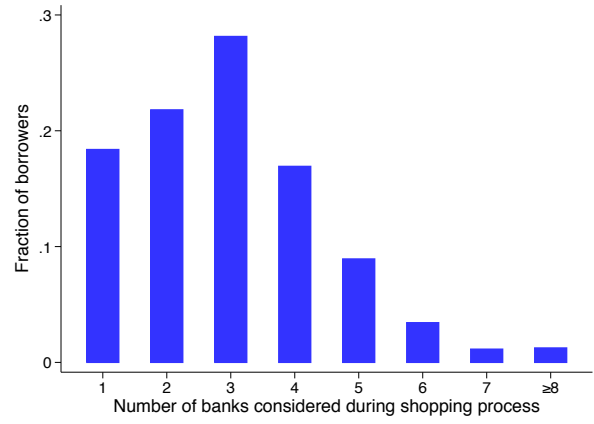


Notes: This figure displays interest rate margins. The red line displays the density of raw interest rate margins in the data. Each additional density displays margins residualized by an increasingly richer sets of covariates, from month FEs to month-bank-size-term-risk FEs.

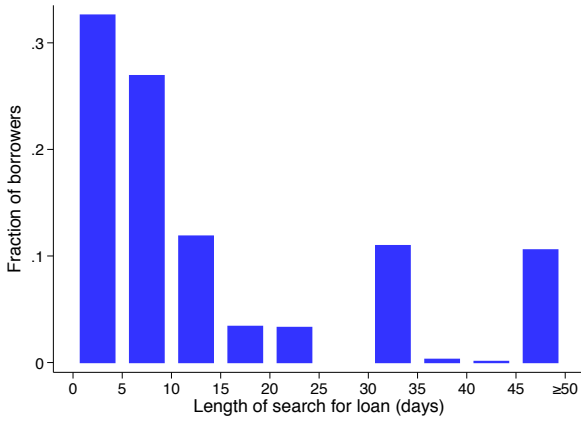
Figure A.8: Survey evidence supporting modeling choices



(a) Perceived price dispersion



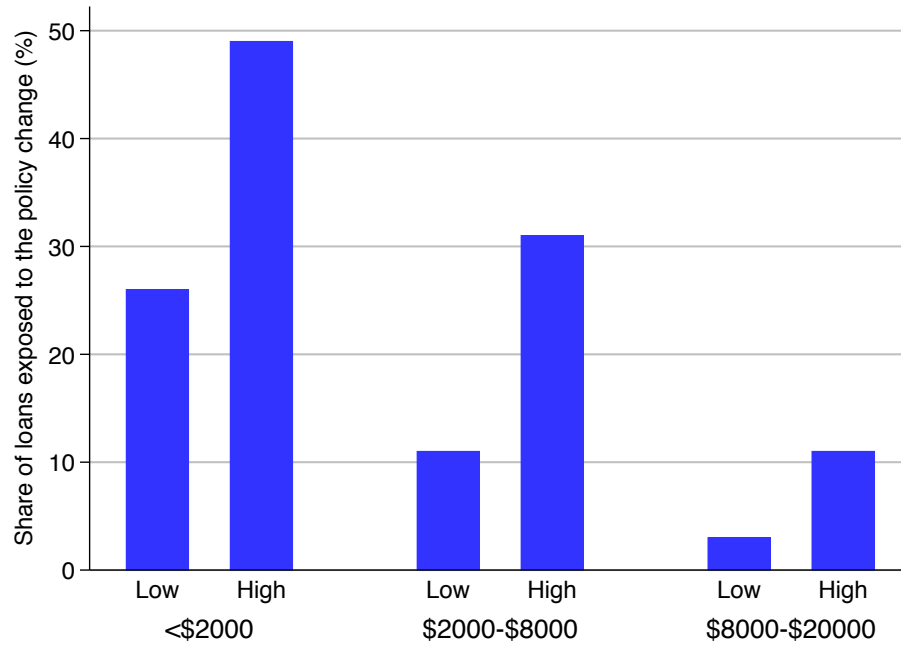
(b) Number of banks considered in shopping process



(c) Length of shopping process

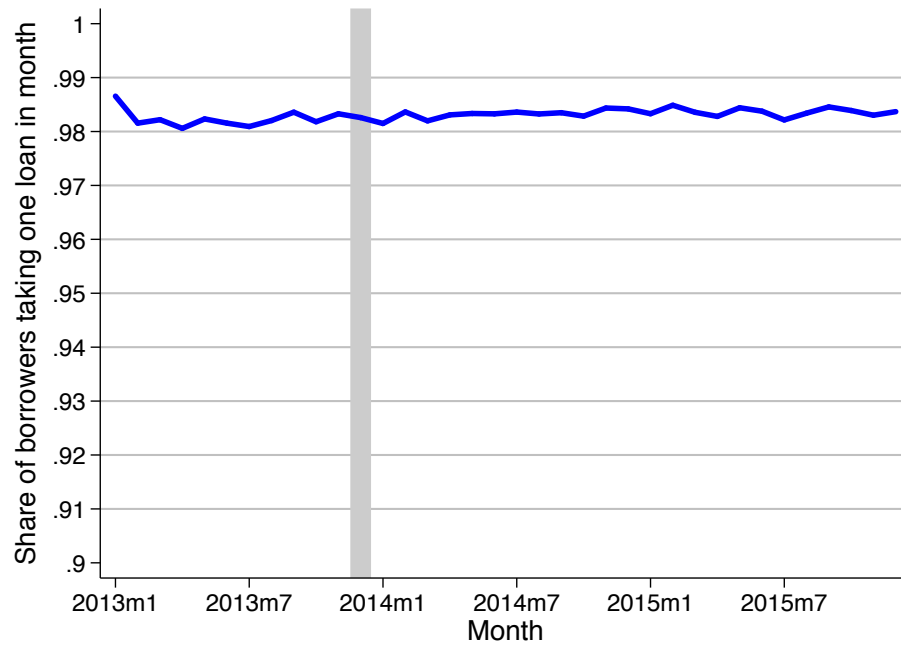
Notes: These figures display survey results related to modeling choices. Panel (a) shows a histogram of the number of banks that borrowers reported to consider during the shopping process for loans. Panel (b) shows a histogram of the length of the search process measured in days between beginning of their search to the end of it, regardless of the approval or rejection of their applications. Panel (c) shows a histogram of perceived price dispersion in the market, as measured by the ratio of the range between the lowest and the highest monthly payment in the market to the highest monthly payment in the market, $\frac{p_H - p_L}{p_H}$.

Figure A.9: Exposure to interest rate regulation by loan size and borrower risk



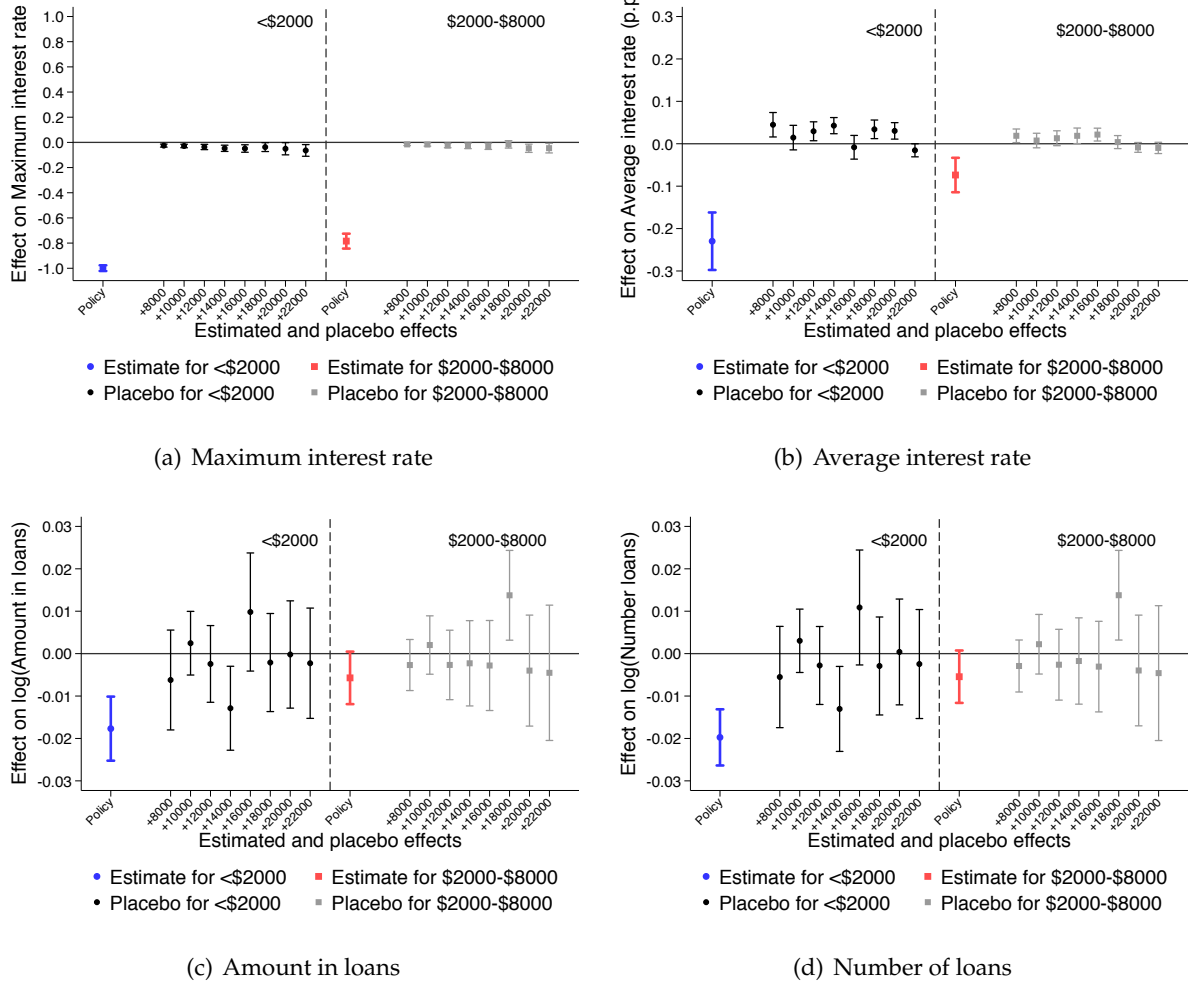
Notes: This figure displays a measure of exposure to interest rate regulation across loan size and borrower risk. Exposure to the policy is calculated as the share of loans that were signed before the policy was implemented in December 2013, at interest rates higher than the interest rate cap once the policy was fully in place in December 2015.

Figure A.10: Number of loans per borrower and month, conditional on borrower



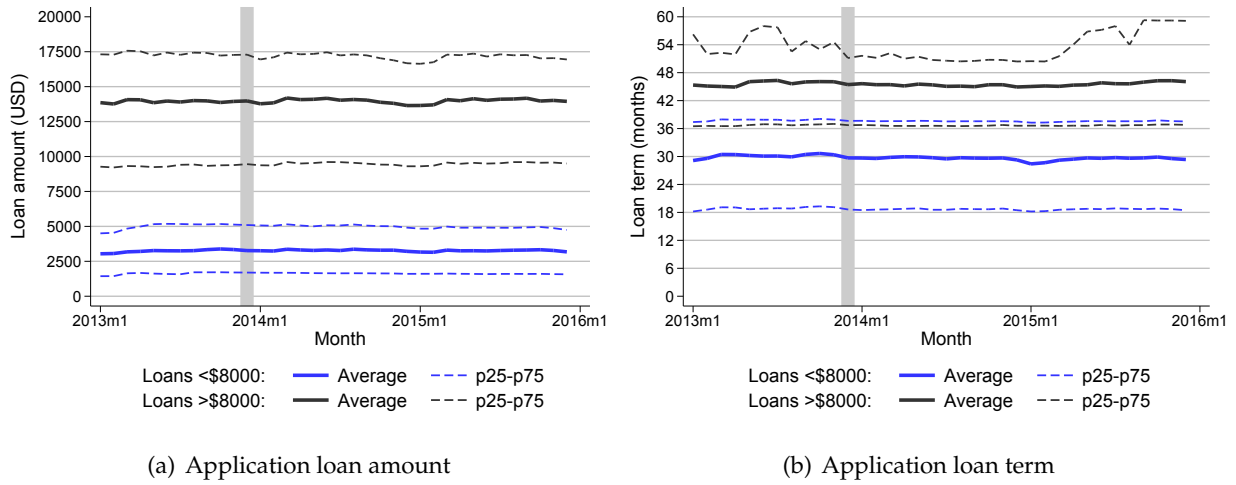
Notes: This figure displays the evolution of the share of borrowers taking only one loan in a month in which they borrow. The gray vertical line indicates the policy change.

Figure A.11: Differences-in-differences effects of placebo policies



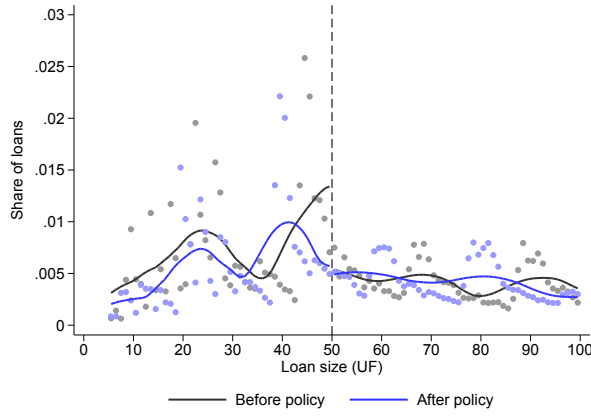
Notes: This figure displays the contrast between our estimates from Table 3 and Table 4 (blue and red) with estimates for a range of placebo policies (black and gray). In each figure, the left panel displays results for loans of $\$0-\$2,000$ and the right panel displays results for loans of $\$2,000-\$8,000$. Placebo policies are constructed by using the same policy intensity variables to estimate effects of interest rate regulation on different parts of the loan size distribution. The first placebo policy adds $\$8,000$ to the actual policy definition, and placebo policies subsequently add $\$2,000$. Each dot indicates the estimated coefficient, while spikes indicate 95% confidence intervals clustered at the risk bin-product bin level. All regressions are weighted by the number of loans in the product type bin-risk bin before the policy was implemented.

Figure A.12: Evolution of distribution of application loan amount and term

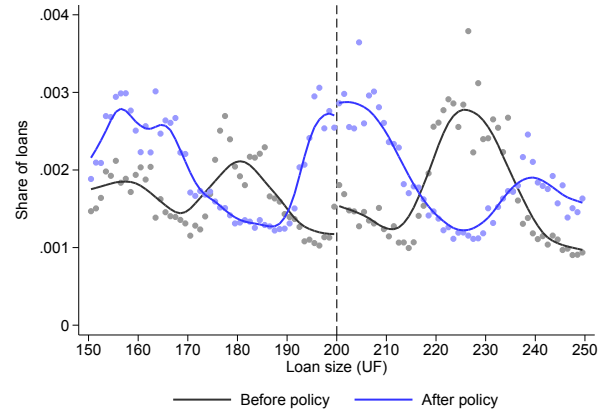


Notes: This figure displays the evolution of the distribution of application loan amount and term separately for loans of \$0-\$8,000 and loans of \$8,000-\$20,000. In particular, each panel displays the average and the 25th and 75th percentiles of each variable for each month.

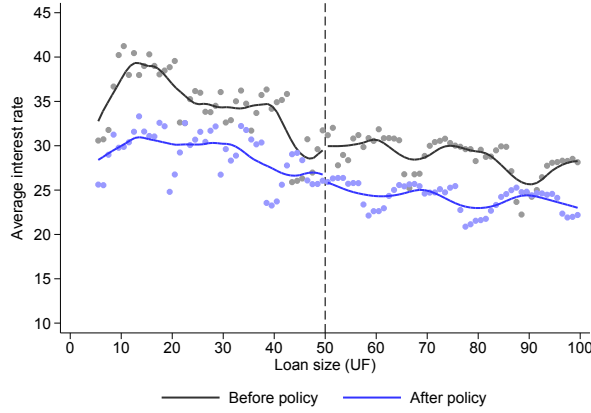
Figure A.13: Distribution of loan size around policy thresholds



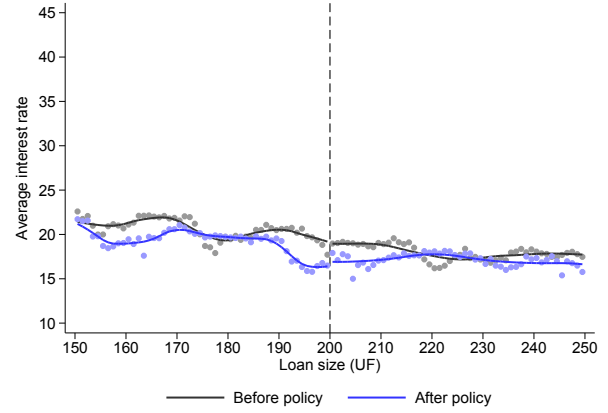
(a) Loan size around \$2,000 threshold, before and after



(b) Loan size around \$8,000 threshold, before and after



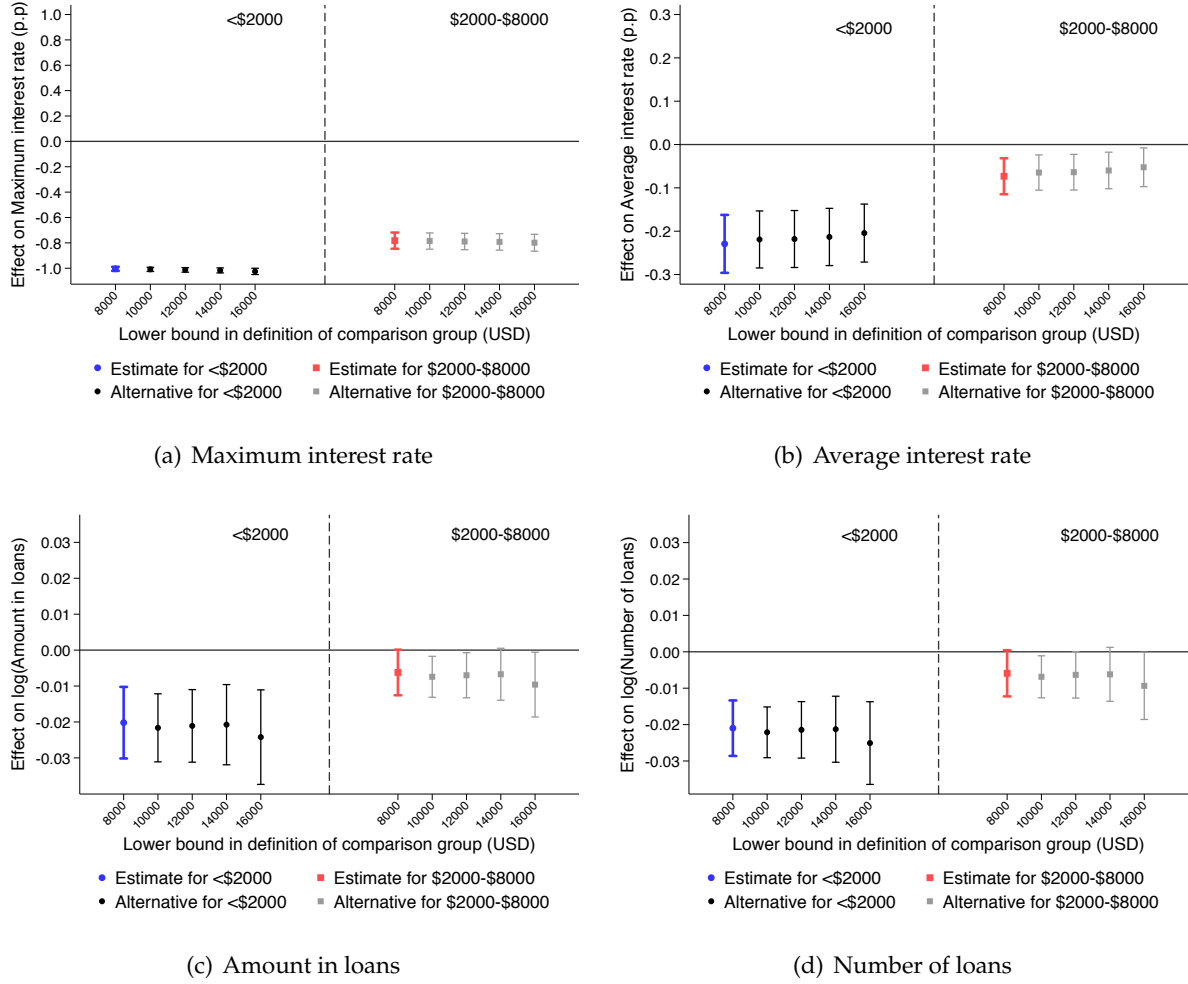
(c) Rates around \$2,000 threshold, before and after



(d) Rates around \$8,000 threshold, before and after

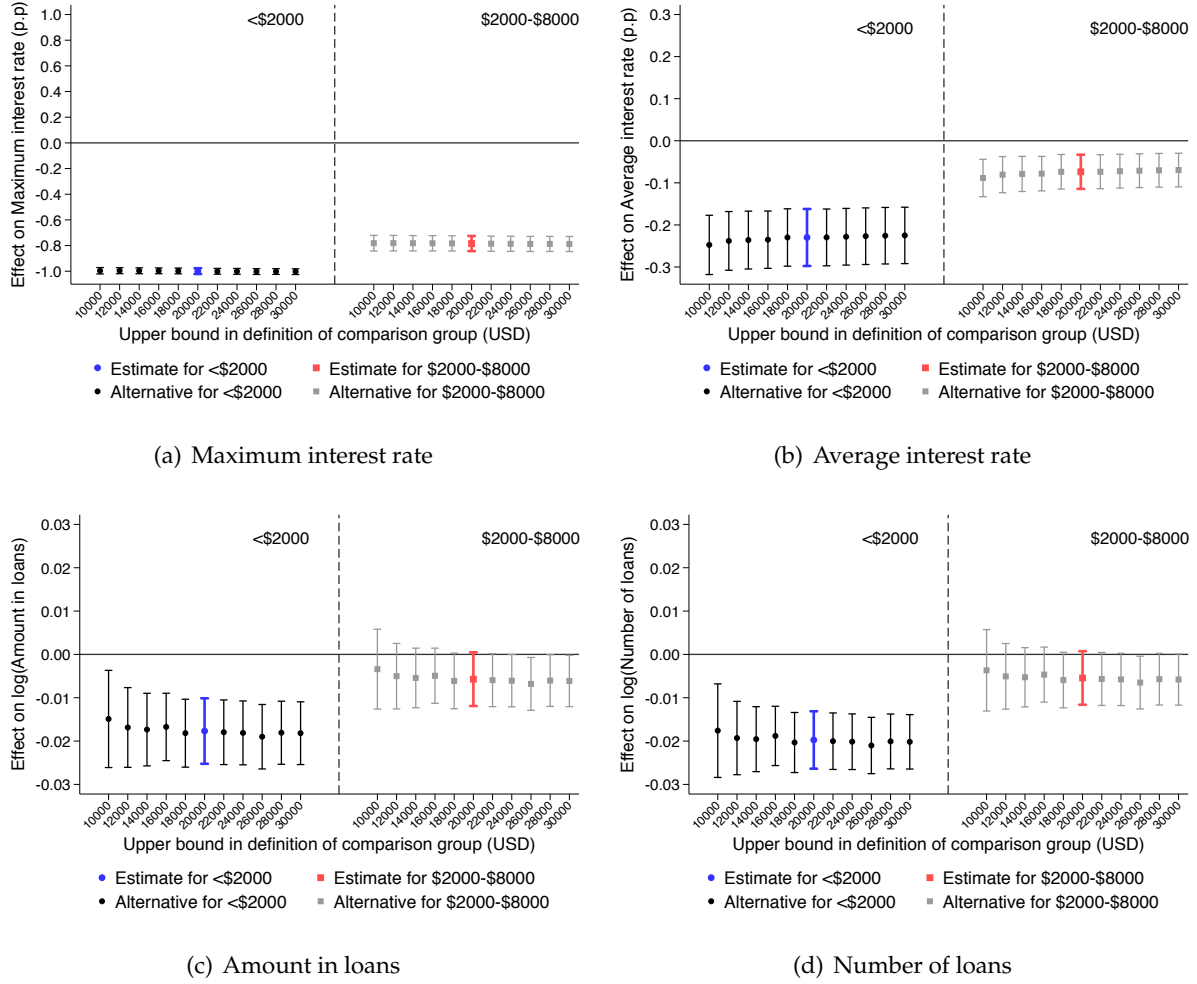
Notes: This figure displays shares of loans and average interest rates by loan size around policy size thresholds at 50UF (\$2,000) and 200UF (\$8,000). The data is binned in bins of 1UF (\$40). For each bin, dots indicate the share of loans originated and average interest rates. Shares are computed across the \$0-\$20,000 interval. Gray dots indicate loan originations in the semester before the policy was implemented, between January 2013 and November 2013. Blue dots indicate loan originations in the last semester before the policy was fully in place, between January 2016 and November 2016. Gray and blue lines are local polynomial fits of the relationship between number of loans and loan size in Panels (a) and (b) and between interest rates and loan size in Panels (c) and (d), allowed to differ at both sides of the relevant policy threshold.

Figure A.14: Differences-in-differences effects under alternative comparison groups



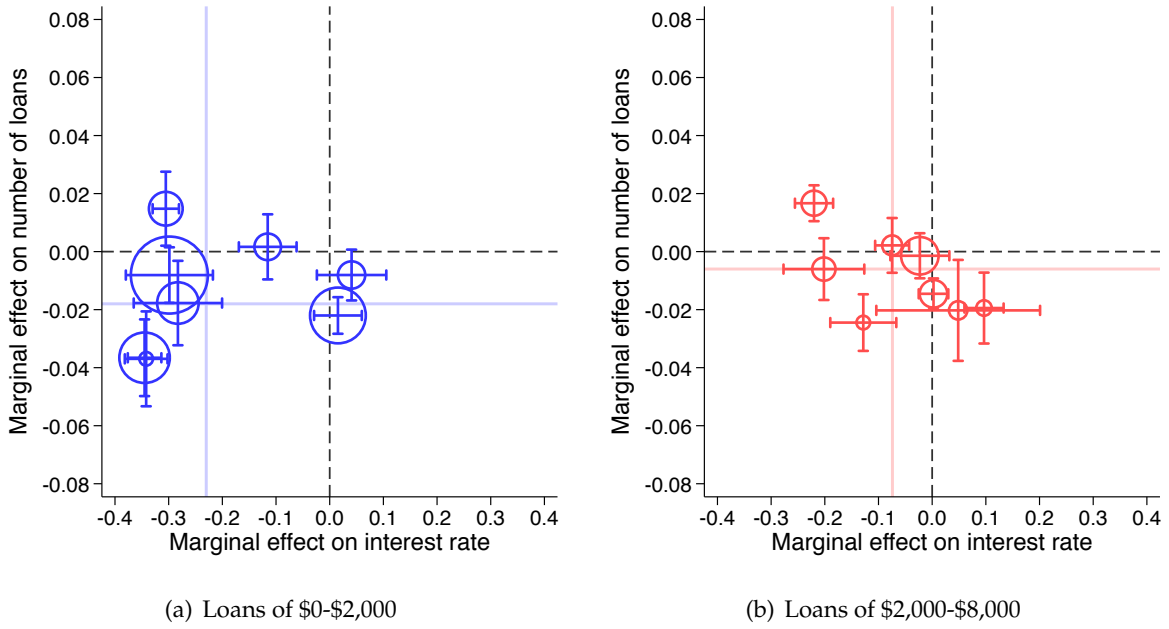
Notes: This figure displays the contrast between our estimates from Table 3 and Table 4 (blue and red) with estimates for a range of alternative comparison groups (black and gray). In each figure, the left panel displays results for loans of \$0-\$2,000 and the left panel displays results for loans of \$2,000-\$8,000. Alternative comparison groups are constructed by shifting the lower bound in the definition of the comparison group, so as to vary the inclusion criterion in terms of the distance to the cutoff set by the policy at \$8,000. The first comparison group sets such upper bound at \$8,000 as in our baseline results, and subsequent alternative comparison groups include loans \$2,000 larger in size. Each dot indicates the estimated coefficient, while spikes indicate 95% confidence intervals clustered at the risk bin-product bin level. All regressions are weighted by the number of loans in the product type bin-risk bin before the policy was implemented.

Figure A.15: Differences-in-differences effects under alternative comparison group size



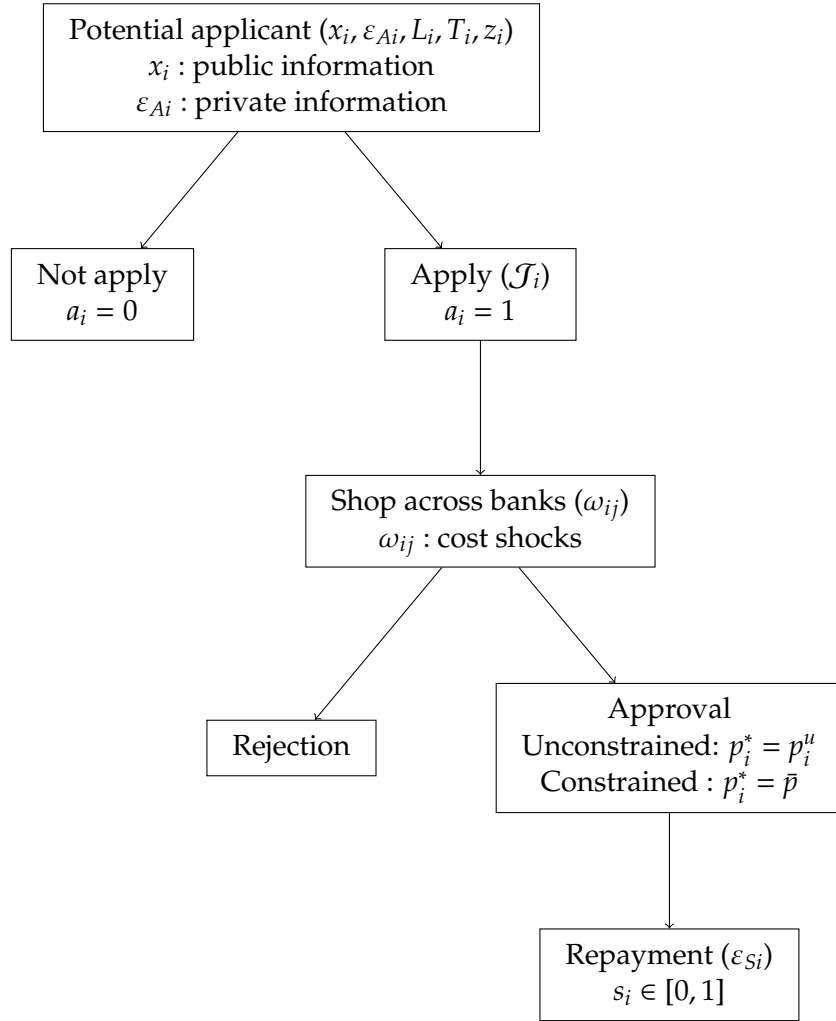
Notes: This figure displays the contrast between our estimates from Table 3 and Table 4 (blue and red) with estimates for a range of alternative comparison groups (black and gray). In each figure, the left panel displays results for loans of \$0-\$2,000 and the left panel displays results for loans of \$2,000-\$8,000. Alternative comparison groups are constructed by shifting the upper bound in the definition of the comparison group, so as to vary the inclusion criterion in terms of the distance to the cutoff set by the policy at \$8,000. The first comparison group sets such upper bound at \$10,000, and subsequent alternative comparison groups include loans \$2,000 larger in size. Each dot indicates the estimated coefficient, while spikes indicate 95% confidence intervals clustered at the risk bin-product bin level. All regressions are weighted by the number of loans in the product type bin-risk bin before the policy was implemented.

Figure A.16: Heterogeneity in effects across banks



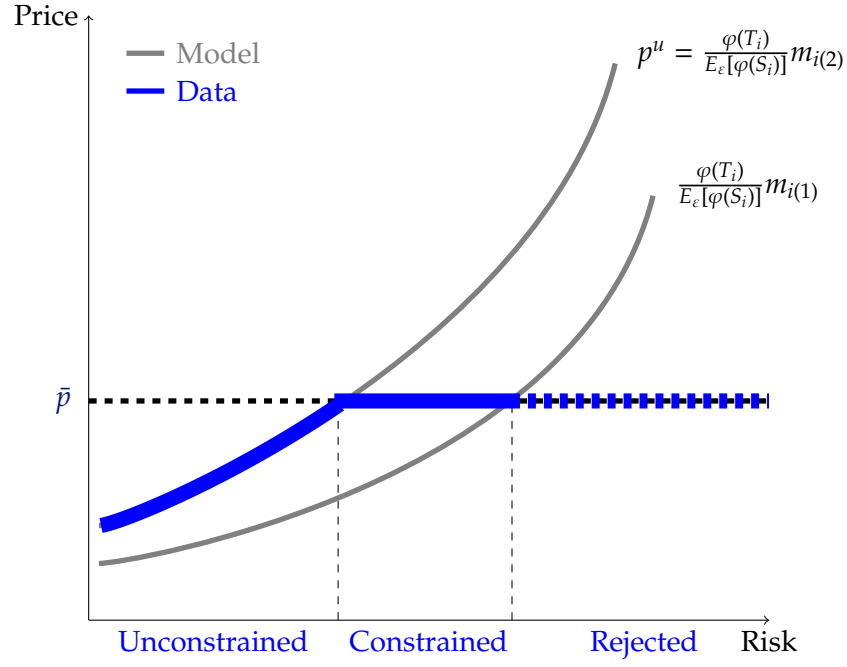
Notes: These figures display marginal effects of interest rate regulation on both number of loans and prices across banks. In each panel, circles indicate estimates for effect on prices on the x-axis and on number of loans on the y-axis the size of the circle is given by the market share of the bank; and spikes indicate standard errors clustered at the risk bin-product bin level. All regressions are weighted by the number of loans in the product type bin-risk bin before the policy was implemented. Solid lines indicate marginal effects estimated across banks, as displayed in Table 3 and Table 4.

Figure A.17: Timing of the model



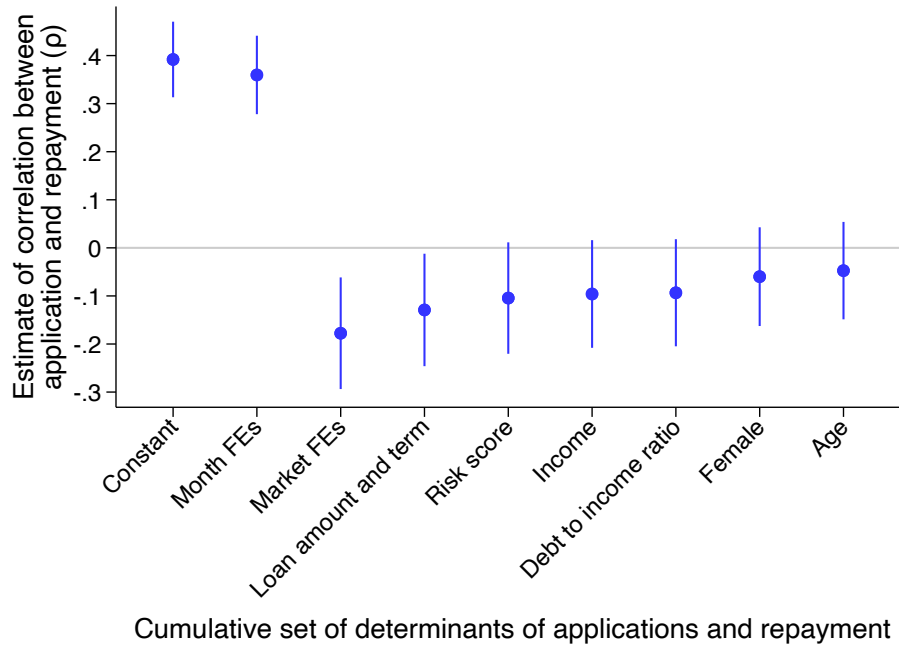
Notes: This figure displays the timing and structure of the model. Exogenous observables characterizing the consumer are covariates x_i , loan amount and term (L_i, T_i) , and application cost shifters z_i . Consumers draw a consideration set $\mathcal{J}_i$ upon application. Consumer unobservables are $(\varepsilon_{Ai}, \varepsilon_{Si})$, whereas bank unobservables are cost shocks ω_{ij} . Finally, endogenous variables are application choices a_i , approval and pricing choices p_i , and repayment s_i .

Figure A.18: Intuition for identification of bank cost



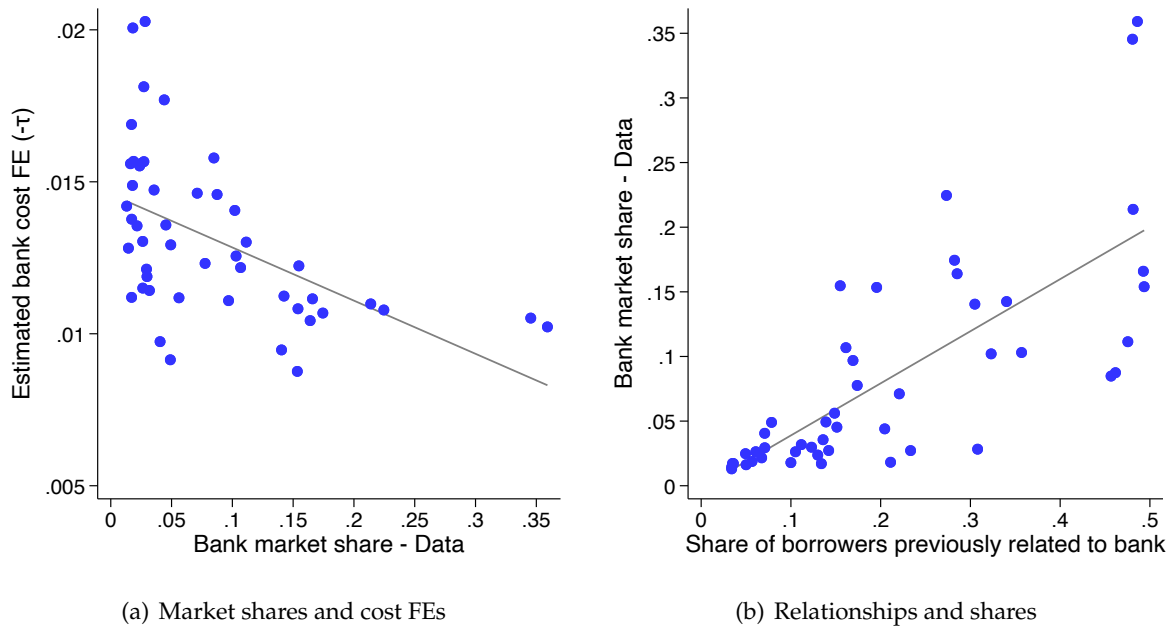
Notes: This figure provides an illustration of how observed outcomes inform the identification of bank cost. The blue line combines observed application outcomes with observed prices, while the gray lines represent the first order statistic of cost and optimal price for the bank with the lowest cost in the market.

Figure A.19: Selection estimate under different specifications



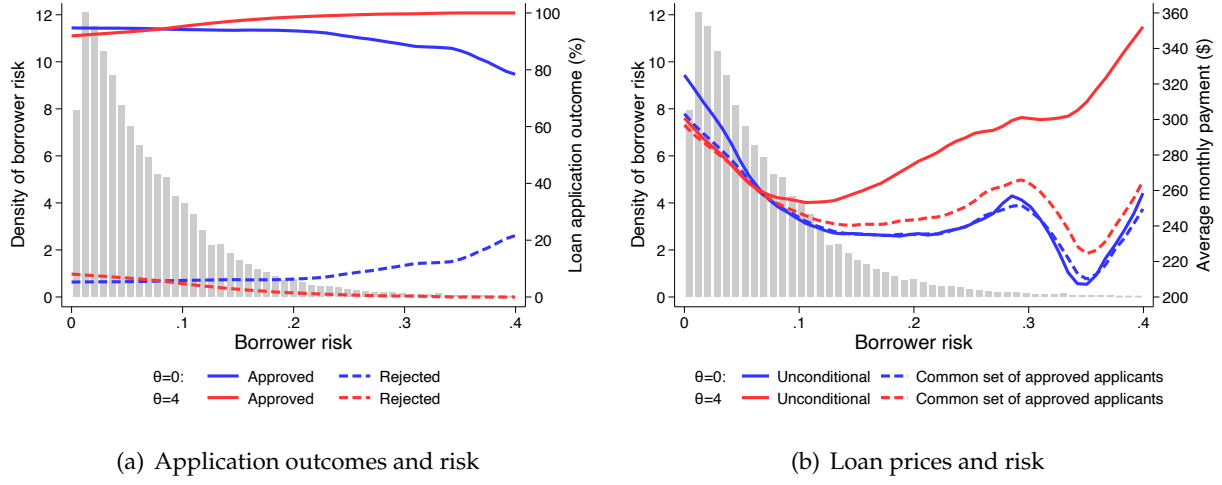
Notes: This figure displays estimates for the coefficient measuring the correlation between shocks to application and repayment (ρ) under different specifications. In particular, we estimate the application and repayment equations using an increasingly rich set of borrower covariates for *both* of them to assess the extent to which available observables are able to capture patterns of risk selection into the market. Dots indicate estimates of ρ . Lines indicate 95% confidence intervals. The first specification we consider includes a constant, and subsequent specifications add more covariates sequentially. Our preferred specification for the analysis in the paper is that with the full vector of covariates.

Figure A.20: Relationship between cost estimates and data



Notes: Panel (a) in this figure displays observed market shares and estimates for bank fixed effects τ_j in ω_{ij} . Panel (b) in this figure displays the correlation between observed shares of previously related borrowers and observed bank market shares.

Figure A.21: Risk-based interest rate caps, additional results



Notes: These figures display simulated market outcomes under different levels of risk-based interest rate regulation given by θ . Baseline outcomes for $\theta = 0$ correspond to simulated equilibrium outcomes for November 2015 under the baseline regulation. Panel (a) displays a local polynomial fit of application outcomes over borrower risk, along with a histogram of borrower risk. Panel (b) displays a local polynomial fit of loan monthly payments over borrower risk, along with a histogram of borrower risk. Solid lines provide results for approved loans. Dashed lines provide results for the common set of approved loans.

Table A.1: Summary statistics for household survey

	N	Mean	SD	p10	p50	p90
Panel A - Experience in credit market						
Debit account	1,003	0.84	0.36	0.00	1.00	1.00
Checking account	1,003	0.89	0.31	0.00	1.00	1.00
Debit card	1,003	0.97	0.17	1.00	1.00	1.00
Credit card	1,003	0.98	0.15	1.00	1.00	1.00
Credit line	1,003	0.88	0.32	0.00	1.00	1.00
Consumer loan	1,003	0.98	0.14	1.00	1.00	1.00
Car loan	1,003	0.35	0.48	0.00	0.00	1.00
Mortgage	1,003	0.50	0.50	0.00	0.00	1.00
College loan	1,003	0.34	3.15	0.00	0.00	1.00
Panel B - Shopping behavior						
Number of considered banks	963	2.99	1.65	1.00	3.00	5.00
Number of applications	963	1.35	0.80	1.00	1.00	2.00
Duration of search period in days	963	15.15	28.65	1.00	7.00	30.00
Offline shopping	963	0.81	0.39	0.00	1.00	1.00
Perceived range of prices (%)	994	25.70	12.64	10.53	24.81	41.18
Panel C - Economic hardships						
Experienced economic hardship	1,003	0.59	0.49	0.00	1.00	1.00
Financed economic hardship with formal credit	1,003	0.55	0.50	0.00	1.00	1.00
Financed economic hardship with savings/assets	1,003	0.27	0.44	0.00	0.00	1.00
Financed economic hardship with other	1,003	0.35	0.48	0.00	0.00	1.00
Cut expenditure on non-durables	1,003	0.87	0.34	0.00	1.00	1.00
Cut expenditure on personal care	1,003	0.37	0.48	0.00	0.00	1.00
Cut expenditure on health	1,003	0.18	0.38	0.00	0.00	1.00
Cut expenditure on education	1,003	0.12	0.32	0.00	0.00	1.00
Cut expenditure on home services	1,003	0.36	0.48	0.00	0.00	1.00
Cut expenditure on transportation	1,003	0.15	0.35	0.00	0.00	1.00
Stopped paying consumer loan	1,003	0.28	0.45	0.00	0.00	1.00
Stopped paying credit card	1,003	0.43	0.50	0.00	0.00	1.00
Stopped paying utility bills	1,003	0.11	0.31	0.00	0.00	1.00
Stopped paying rent	1,003	0.02	0.14	0.00	0.00	0.00
Stopped paying mortgage	1,003	0.06	0.24	0.00	0.00	0.00
Stopped paying car loan	1,003	0.07	0.25	0.00	0.00	0.00
Stopped paying student loan	1,003	0.06	0.23	0.00	0.00	0.00
Stopped paying health bills	1,003	0.07	0.25	0.00	0.00	0.00
Stopped paying other bills	1,003	0.13	0.34	0.00	0.00	1.00
Panel D - Borrower attributes						
Age	1,003	42.18	8.80	32.00	41.00	55.00
Female	1,003	0.39	0.49	0.00	0.00	1.00
Approval probability	981	0.59	0.07	0.49	0.60	0.67
Annual income	981	17,557.88	7,803.02	7,772.14	16,853.41	28,616.13
Financial literacy score (1-3)	1,003	1.75	0.81	1.00	2.00	3.00

Notes: This table displays summary statistics for our household survey.

Table A.2: Determinants of loan performance

	(1)	(2)	(3)	(4)	(5)
	1{Default}				
log(Income)	-0.421*** (0.004)	-0.376*** (0.005)	-0.501*** (0.005)	-0.498*** (0.005)	-0.498*** (0.005)
Consumer debt to income ratio	0.013*** (0.004)	-0.020*** (0.006)	0.054*** (0.006)	0.054*** (0.006)	0.054*** (0.006)
Mortgage debt to income ratio	-0.217*** (0.005)	0.136*** (0.010)	0.101*** (0.010)	0.104*** (0.010)	0.104*** (0.010)
log(Consumer debt)		0.108*** (0.011)	0.111*** (0.011)	0.109*** (0.011)	0.108*** (0.011)
No consumer debt		1.059*** (0.029)	0.934*** (0.029)	0.747*** (0.031)	0.746*** (0.031)
No consumer debt ≥90-day default		-0.531*** (0.017)	-0.414*** (0.017)	-0.414*** (0.017)	-0.415*** (0.017)
No consumer debt ≥90-day default		-0.458*** (0.008)	-0.467*** (0.008)	-0.468*** (0.008)	-0.468*** (0.008)
Consumer ≥90-day default to debt ratio		0.591*** (0.035)	0.630*** (0.036)	0.618*** (0.036)	0.619*** (0.036)
Consumer <90-day default to debt ratio		0.425*** (0.093)	0.554*** (0.094)	0.519*** (0.094)	0.518*** (0.094)
log(Mortgage debt)		-0.746*** (0.040)	-0.794*** (0.041)	-0.821*** (0.041)	-0.822*** (0.041)
No mortgage debt		-0.663*** (0.077)	-0.838*** (0.080)	-0.911*** (0.080)	-0.912*** (0.080)
No mortgage debt ≥90-day default		-0.308*** (0.049)	-0.339*** (0.050)	-0.360*** (0.050)	-0.360*** (0.050)
No mortgage debt <90-day default		-0.626*** (0.025)	-0.620*** (0.025)	-0.625*** (0.025)	-0.625*** (0.025)
Mortgage ≥90-day default to debt ratio		0.052 (0.136)	0.178 (0.136)	0.195 (0.136)	0.195 (0.136)
Mortgage <90-day default to debt ratio		-2.557*** (0.400)	-2.063*** (0.376)	-2.116*** (0.376)	-2.117*** (0.376)
Change in consumer debt		0.226*** (0.005)	0.201*** (0.005)	0.201*** (0.005)	0.201*** (0.005)
Change in consumer debt ≥90d default		0.010*** (0.004)	0.006* (0.004)	0.006* (0.004)	0.006* (0.004)
Change in mortgage debt		-0.011*** (0.003)	-0.012*** (0.003)	-0.012*** (0.003)	-0.012*** (0.003)
Change in mortgage debt ≥90d default		0.042*** (0.006)	0.027*** (0.007)	0.025*** (0.007)	0.025*** (0.007)
Age			-0.411*** (0.003)	-0.407*** (0.003)	-0.407*** (0.003)
Female			-0.308*** (0.007)	-0.310*** (0.007)	-0.309*** (0.007)
Previously related to any bank				-0.283*** (0.016)	-0.283*** (0.016)
Local unemployment rate					0.015** (0.007)
Constant	-1.907*** (0.024)	0.113 (0.078)	0.234*** (0.080)	0.599*** (0.082)	0.603*** (0.082)
Observations	916,934	916,934	916,436	916,436	916,436
Pseudo R-squared	0.034	0.054	0.079	0.080	0.080
Market FE	Y	Y	Y	Y	Y

Notes: All columns display results from logit regressions of individual loan default outcomes on borrower covariates. All covariates are standardized. Credit history variables are computed as average over the year previous to each loan. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.3: Effects excluding loans around the \$8,000 threshold

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Maximum interest rate			Average interest rate			log(Applications)			log(Credit volume)		
	All	Low-risk	High-risk	All	Low-risk	High-risk	All	Low-risk	High-risk	All	Low-risk	High-risk
Loans in \$0 – \$2000	-1.003*** (0.012)	-0.907*** (0.062)	-0.996*** (0.018)	-0.220*** (0.034)	-0.108*** (0.015)	-0.253*** (0.042)	-0.005 (0.004)	0.002 (0.003)	-0.013* (0.007)	-0.019*** (0.003)	-0.013*** (0.003)	-0.029*** (0.004)
Loans in \$2000 – \$8000	-0.838*** (0.020)	-0.662*** (0.028)	-0.831*** (0.027)	-0.075*** (0.024)	-0.032*** (0.011)	-0.120*** (0.028)	-0.001 (0.004)	0.001 (0.003)	-0.007 (0.006)	-0.008** (0.003)	-0.007** (0.003)	-0.014*** (0.003)
Observations	2,304	2,304	2,145	2,304	2,304	2,145	2,304	2,304	2,034	2,304	2,304	2,304
R-squared	0.991	0.979	0.985	0.983	0.986	0.964	0.995	0.990	0.994	0.984	0.984	0.966
	log(Number of loans)			Risk-I	Risk-H	log(Income)	Share of loans under 1Y default			Months under 1Y default		
	All	Low-risk	High-risk	All	All	All	All	Low-risk	High-risk	All	Low-risk	High-risk
Loans in \$0 – \$2000	-0.021*** (0.003)	-0.012*** (0.003)	-0.032*** (0.003)	-0.082*** (0.005)	-0.034*** (0.011)	0.004*** (0.001)	-0.097*** (0.018)	-0.066*** (0.007)	-0.112*** (0.036)	-0.004*** (0.001)	-0.002*** (0.000)	-0.003*** (0.001)
Loans in \$2000 – \$8000	-0.008** (0.003)	-0.006* (0.003)	-0.015*** (0.003)	-0.027*** (0.006)	-0.009 (0.008)	0.000 (0.001)	-0.050*** (0.012)	-0.023*** (0.007)	-0.080*** (0.025)	-0.002*** (0.000)	-0.001*** (0.000)	-0.003*** (0.001)
Observations	2,304	2,304	2,304	2,304	2,304	2,304	2,304	2,304	2,145	2,304	2,304	2,145
R-squared	0.992	0.988	0.987	0.988	0.978	0.989	0.909	0.720	0.696	0.907	0.698	0.700
Product bin FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Product bin-month of year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes: This table display estimated coefficients from equation (4) using a sample that excludes loans of size between \$6,000 and \$10,000. For each outcome, the regression is estimated across borrower risk bins and separately by borrower risk bin. All regressions include risk bin-product bin fixed effects and risk bin-month fixed effects. For default outcomes, the estimating sample is restricted to loans originated before December 2015, so as to allow for a year long period after origination in which the outcomes are measured. Share of loans under default in first year is computed in a 0-100 scale. Predicted risk is computed as described in Section 2.2.3 and measured in a 0-100 scale. Income is measured in thousands of U.S. dollars. All regressions are weighted by the number of loans in the product type bin-risk bin before the policy was implemented. Clustered standard errors at the risk bin-product bin level are displayed in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.4: Market power and interest rate regulation

	(1)	(2)	(3)	(4)	(5)	(6)
	Δ Interest	Size of consideration set $E[\mathcal{J}_i]$				
Outcome	rate cap	1.2	2.4	3.6	4.8	6
Δ Number of loans (%)	-5.00	0.86	0.02	-0.48	-0.46	-0.42
	-10.00	1.62	-0.29	-1.89	-1.88	-1.70
	-15.00	2.45	-1.46	-5.30	-6.24	-5.72
	-20.00	3.21	-4.68	-13.48	-15.87	-16.17
	-25.00	4.13	-12.05	-27.46	-33.15	-34.22
	-30.00	4.80	-25.07	-47.43	-54.54	-56.13
Δ Number of applications (%)	-5.00	0.86	0.04	-0.38	-0.42	-0.32
	-10.00	1.62	-0.17	-1.46	-1.52	-1.38
	-15.00	2.45	-1.08	-4.31	-5.13	-4.85
	-20.00	3.21	-3.75	-11.33	-13.43	-13.53
	-25.00	4.13	-9.85	-23.47	-28.61	-29.38
	-30.00	4.81	-21.17	-41.56	-48.17	-49.47
Δ Average monthly payment (\$)	-5.00	-9.68	-2.46	-0.92	-0.44	-0.28
	-10.00	-19.56	-5.07	-2.28	-1.33	-1.13
	-15.00	-29.30	-8.23	-4.44	-3.69	-3.18
	-20.00	-39.08	-11.82	-8.15	-7.52	-7.13
	-25.00	-48.74	-16.23	-13.04	-12.38	-13.13
	-30.00	-58.37	-21.35	-18.98	-19.49	-20.03
Δ Average consumer surplus (\$)	-5.00	2.33	0.18	-1.22	-1.30	-1.05
	-10.00	4.65	-0.53	-4.97	-5.51	-4.89
	-15.00	6.96	-3.71	-14.73	-17.56	-16.83
	-20.00	9.24	-12.58	-36.57	-45.14	-45.73
	-25.00	11.39	-32.10	-75.70	-92.61	-95.61
	-30.00	13.15	-66.95	-129.49	-151.63	-156.81
Δ Average profit (\$)	-5.00	-1.83	-0.53	-0.17	-0.05	-0.03
	-10.00	-3.72	-1.09	-0.41	-0.19	-0.14
	-15.00	-5.59	-1.69	-0.85	-0.58	-0.48
	-20.00	-7.52	-2.46	-1.67	-1.37	-1.25
	-25.00	-9.41	-3.53	-2.86	-2.68	-2.53
	-30.00	-11.37	-4.99	-4.46	-4.32	-4.08

Notes: This table displays results for simulated effects of reforms to interest rate regulation for different degrees of market power. To vary market power, we adjust consideration probabilities and thus the size of borrower consideration sets. We provide results for $k = 5$ competitive environments. To adjust consideration probabilities, let $\phi_{ij} = \frac{\exp(x_{ij})}{1 + \exp(x_{ij})}$, from where we can solve for x_{ij} . Then we find values for λ^k in $\tilde{\phi}_{ij}^k = \frac{\exp(x_{ij})}{\exp(1/\lambda^k) + \exp(x_{ij})}$ such that $E[\tilde{\phi}_{ij}^k]$ is a particular value of interest in $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. This scaling assure that counterfactual probabilities are well defined. The average borrower has a consideration set of size $E[|\mathcal{J}_i|] = E[\tilde{\phi}_{ij}^k] \times \mathcal{J}$, with $\mathcal{J} = 12$ in our empirical model. Column (1) displays the reform under study, which are decreases of between 5 and 30 p.p. Column (2) through (6) display the effects of such reform on a particular outcome of interest indicated in the left column, and given a particular market structure indicated in the top row.